Q: Check Similarity of matrices is equivalence rel?

A:-

$A = I^{-1} A I$

$\Rightarrow A \approx A$ (Reflexive)

Let $A \approx B$

$\Rightarrow B = P^{-1} A P$

$\Rightarrow P B P^{-1} = P P^{-1} A P P^{-1}$

$\Rightarrow (P^{-1})^{-1} B P^{-1} = A$

$\Rightarrow B \approx A$ (symmetric)

Let $A \approx B$, $B \approx C$

$\Rightarrow B = P^{-1} A P$, $C = T^{-1} B T$

$C = T^{-1} P^{-1} A P T$

$= (PT)^{-1} A (PT)$

$\Rightarrow A \approx C$ (Transitive)

$\therefore$ Similarity of matrices is an equivalence relation.

Q:- If $B \approx A$ then show that, for any polynomial $f(t)$, $f(B) \approx f(A)$

*Prove that if $A \approx B$, then $B^n = P^{-1} A^n P$

$B = P^{-1} A P$

$\Rightarrow B^2 = (P^{-1} A P)^2$

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